STAT3888: Causal Inference Analysis

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In this document, we will aim to perform some causal inference on our current models, which involves a closer analysis of the variable relationships. In particular, we can assume that `WAIST` is a *confounder* rather than a mediator, so we will likely keep our main models. As this has been confirmed now, I believe we can leave the models as is.

The causal role of `DSMKHHRB` is less certain. Household smoking may plausibly influence liver biomarkers through passive smoke exposure and may also proxy broader household or socioeconomic conditions. However, its direct relationship with meeting physical-activity guidelines is uncertain. We therefore retain it in the primary DAG/model, consistent with the original research question, and evaluate the robustness of the estimated exercise effect in a sensitivity analysis excluding `DSMKHHRB.`

Directed Acyclic Graph (DAG) of Our Variables

▼ Code

```
# DAG Plot
exercise_dag2 = dagitty("
dag {
  EXREGUIC -> ALTGGT

  AGEB -> EXREGUIC
  AGEB -> ALTGGT

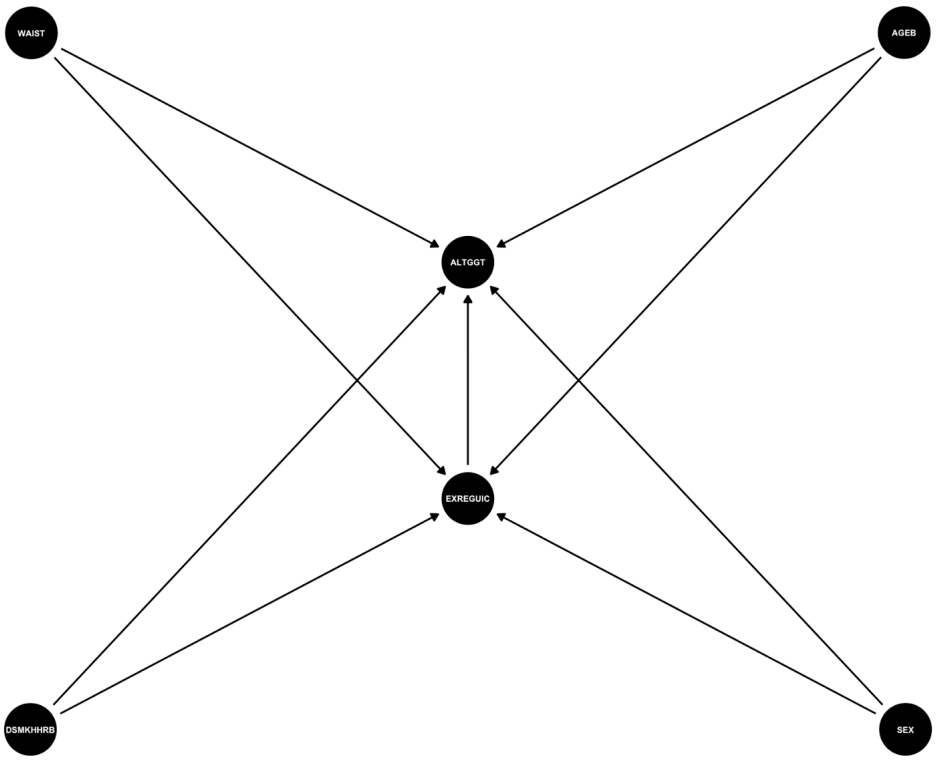
  SEX -> EXREGUIC
  SEX -> ALTGGT

  WAIST -> EXREGUIC
  WAIST -> ALTGGT

  DSMKHHRB -> EXREGUIC
  DSMKHHRB -> ALTGGT
}
")

ggdag(exercise_dag2, text_size = 2) +
```

```
  theme_dag()
```



▼ Code

```
adjustmentSets(
  exercise_dag2,
  exposure = "EXREGUIC",
  outcome = "ALTGGT"
)
```

```
{ AGEB, DSMKHHRB, SEX, WAIST }
```

The adjustment set is given by $\{\texttt{AGEB}, \texttt{DSMKHHRB}, \texttt{SEX}, \texttt{WAIST}\}$. This is the set of covariates to include in the model, which is what we had initially as well. On this note, I don't think doing sensitivity analysis where (with and without `WAIST`) is necessary.

Marginal Effects

Before we check marginal effects and overall effect plots as well, we will load the data sets and our models.

▼ Code

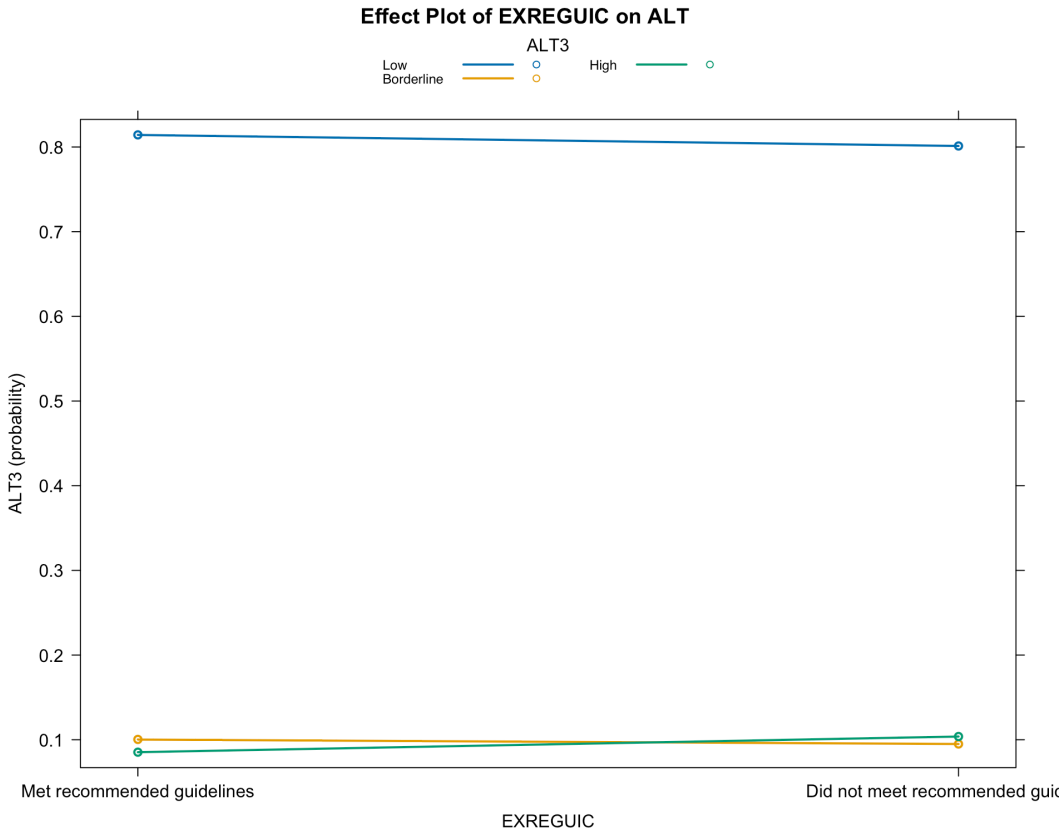
```
# Load data sets
pdat.full = readRDS(here("Objects", "pdat_full.rds"))
fdat.full = readRDS(here("Objects", "fdat_full.rds"))

# Load models
alt.mlr = readRDS(here("Objects", "MLR_ALT3.rds"))
ggt.mlr = readRDS(here("Objects", "MLR_GGT3.rds"))
alt.fmlr = readRDS(here("Objects", "Full_data_MLR_ALT3.rds"))
ggt.fmlr = readRDS(here("Objects", "Full_data_MLR_GGT3.rds"))
```

Then, we use the `effects` package to make effects plots for the models in the full population (since it may be possible that Q2 will focus on the unrestricted population `fdat.full`).

▼ Code

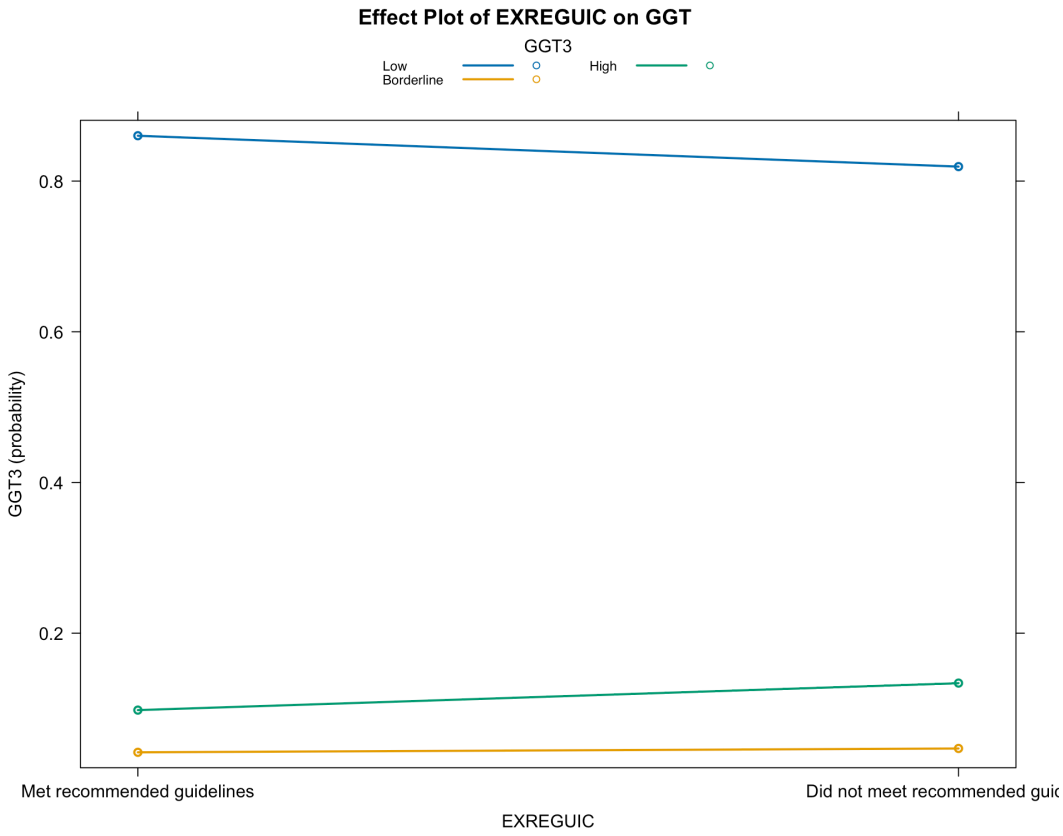
```
# Effects
eff.alt = effect("EXREGUIC", alt.fmlr)
plot(eff.alt, multiline = TRUE, main = "Effect Plot of EXREGUIC
```



▼ Code

```
eff.ggt = effect("EXREGUIC", ggt.fmlr)
```

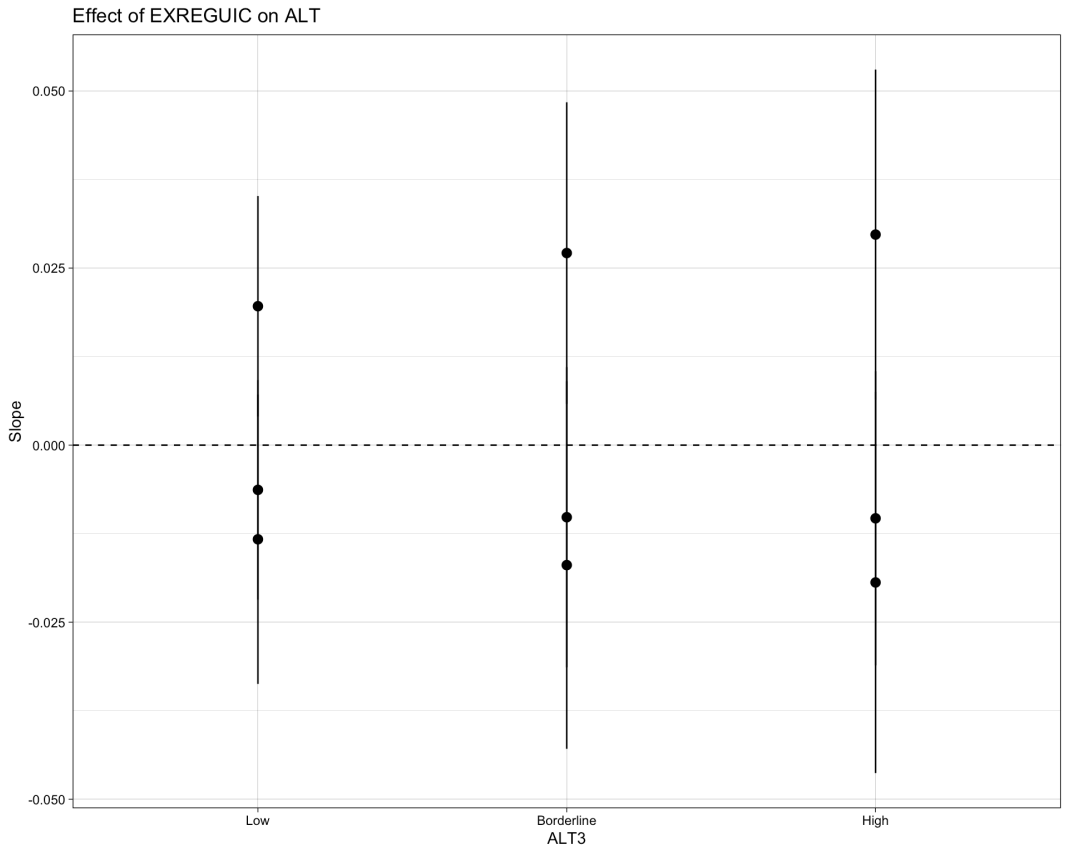
```
plot(eff.ggt, multiline = TRUE, main = "Effect Plot of EXREGUIC
```



Next, we find the marginal effects using `marginaleffects::slopes()` which are a better interpretation than our log odds.

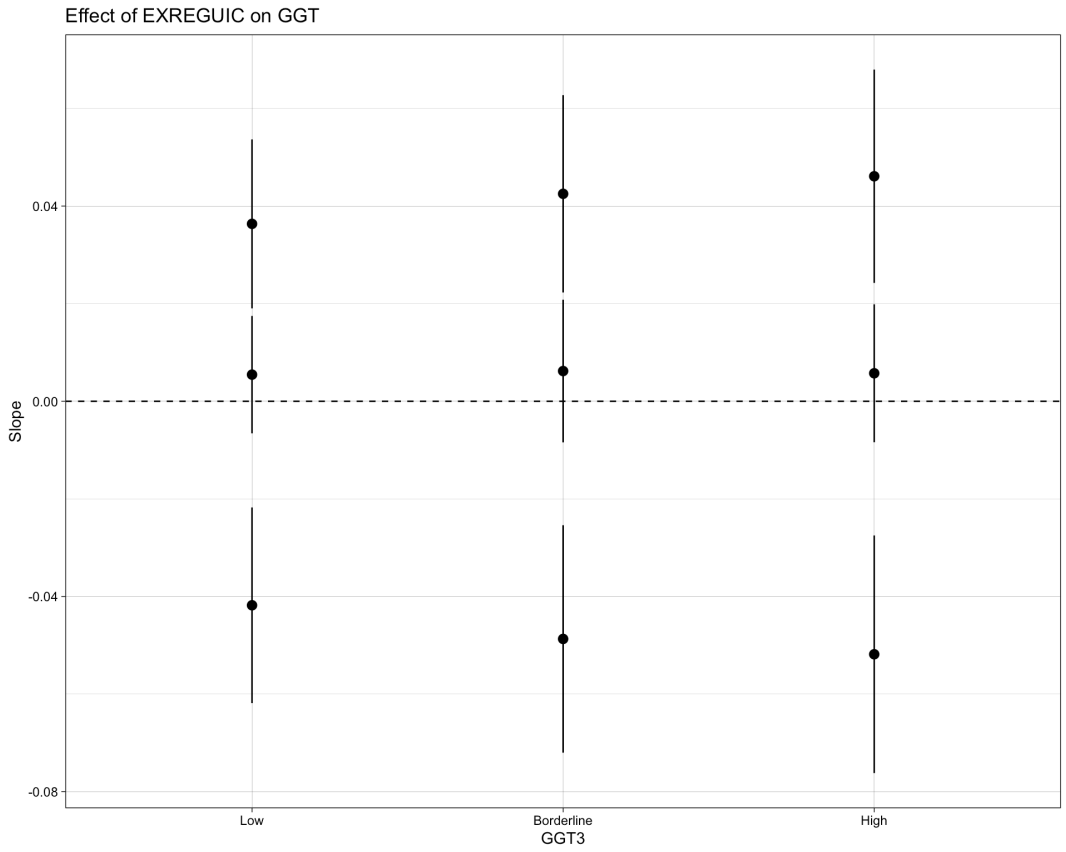
▼ Code

```
# Slopes
plot_slopes(alt.fmlr, variables = "EXREGUIC", by = "ALT3") +
  geom_hline(yintercept = 0, linetype = "dashed", color = "blacl
  theme_linedraw() + labs(title = "Effect of EXREGUIC on ALT")
```



▼ Code

```
plot_slopes(ggt.fmlr, variables = "EXREGUIC", by = "GGT3") +
  geom_hline(yintercept = 0, linetype = "dashed", color = "blacl
  theme_linedraw() + labs(title = "Effect of EXREGUIC on GGT")
```



▼ Code

```
# Average Comparisons
alt.comp = avg_comparisons(alt.fmlr, variables = "EXREGUIC", typ
alt.comp %>% tidy() %>% select(-c(contrast)) %>%
  kable(caption = "Average Comparisons of ALT Using MLR (Contras
        digits = 3, col.names = c("Term", "Group", "Estimate", "
                      "p-value", "s-value", "CI LB", "CI UB"))
```

Average Comparisons of ALT Using MLR (Contrast = Did Not Meet ~ Met)

Term	Group	Estimate	SE	Statistic	p-value	s-value	CI LB	CI UB
EXREGUIC	Low	-0.014	0.011	-1.297	0.195	2.360	-0.036	0.007
EXREGUIC	Borderline	-0.007	0.009	-0.846	0.398	1.330	-0.024	0.010
EXREGUIC	High	0.022	0.009	2.480	0.013	6.252	0.005	0.039

▼ Code

```
ggt.comp = avg_comparisons(ggt.fmlr, variables = "EXREGUIC", typ
ggt.comp%>% tidy() %>% select(-c(contrast)) %>%
  kable(caption = "Average Comparisons of GGT Using MLR (Contras
        digits = 3, col.names = c("Term", "Group", "Estimate", "
```

```
                          "p-value", "s-value", "CI LB", "CI UB"))
```

Average Comparisons of GGT Using MLR (Contrast = Did Not Meet ~ Met)

Term	Group	Estimate	SE	Statistic	p-value	s-value	CI LB	CI UB
EXREGUIC	Low	-0.044	0.011	-4.102	0.000	14.576	-0.064	-0.023
EXREGUIC	Borderline	0.006	0.006	0.871	0.384	1.382	-0.007	0.018
EXREGUIC	High	0.038	0.009	4.120	0.000	14.687	0.020	0.056

The model-based probability plots indicate a generally favourable shift in liver biomarker categories among participants meeting the physical-activity guidelines. For both ALT and GGT, meeting the guideline is associated with a higher predicted probability of the Low category and a lower predicted probability of the High category, with the difference appearing more pronounced for GGT. These plots provide a more intuitive representation of the association than the underlying multinomial log-odds coefficients.

The average comparisons basically answer “On average, how much does the predicted probability of each ALT/GGT category change between meeting and not meeting the guideline?”. In this case, for ALT meeting physical activity guidelines is associated with a 1.442% higher probability of Low ALT, and 2.164% lower probability of High ALT. However, only High ALT results are significant here so our inference must be a bit more careful…

Propensity Scores

We can use `WeightIt` for propensity scoring, and select the GLM option for now; we may make a more extensive analysis later on if this is worth pursuing.

Balance tables and Love plots are diagnostic tools used in statistics and causal inference to evaluate whether confounding covariates are balanced between treatment and control groups after methods like propensity score matching or weighting.

ALT

▼ Code

```
# Formula for ALT; use full population (and complete cases only)
fm.alt = EXREGUIC ~ AGEB + SEX + WAIST + DSMKHHRB
fdat.comp = fdat.full %>% drop_na()
```

```
# WeightIt
wmultinom.alt = weightit(fm.alt, fdat.comp, method = "glm", est
set.cobalt.options(binary = "std")

# Check SMD; generally < 0.1 is considered well-balanced. Also l
bal.tab(wmultinom.alt, thresholds = c(m = .1))
```

```
Balance Measures
                                    Type Diff.Adj
M.Threshold
prop.score                      Distance  -0.0022 Balanced,
<0.1
AGEB_18-19 years                  Binary   0.0018 Balanced,
<0.1
AGEB_20-24 years                  Binary   0.0057 Balanced,
<0.1
AGEB_25-29 years                  Binary  -0.0015 Balanced,
<0.1
AGEB_30-34 years                  Binary  -0.0016 Balanced,
<0.1
AGEB_35-39 years                  Binary  -0.0019 Balanced,
<0.1
AGEB_40-44 years                  Binary   0.0012 Balanced,
<0.1
AGEB_45-49 years                  Binary  -0.0010 Balanced,
<0.1
AGEB_50-54 years                  Binary   0.0006 Balanced,
<0.1
AGEB_55-59 years                  Binary   0.0029 Balanced,
<0.1
AGEB_60-64 years                  Binary  -0.0019 Balanced,
<0.1
AGEB_65-69 years                  Binary  -0.0004 Balanced,
<0.1
AGEB_70-74 years                  Binary   0.0008 Balanced,
<0.1
AGEB_75-79 years                  Binary  -0.0013 Balanced,
<0.1
AGEB_80-84 years                  Binary   0.0007 Balanced,
<0.1
AGEB_85 years and older           Binary  -0.0026 Balanced,
<0.1
SEX_Male                          Binary  -0.0007 Balanced,
<0.1
WAIST_Increased risk              Binary  -0.0001 Balanced,
<0.1
```

```
WAIST_Not at risk                    Binary   0.0023 Balanced,
<0.1
WAIST_Substantially increased risk   Binary  -0.0020 Balanced,
<0.1
DSMKHHRB_None                        Binary   0.0008 Balanced,
<0.1
DSMKHHRB_One                         Binary   0.0030 Balanced,
<0.1
DSMKHHRB_Two                         Binary  -0.0065 Balanced,
<0.1
DSMKHHRB_Three or more               Binary  -0.0020 Balanced,
<0.1

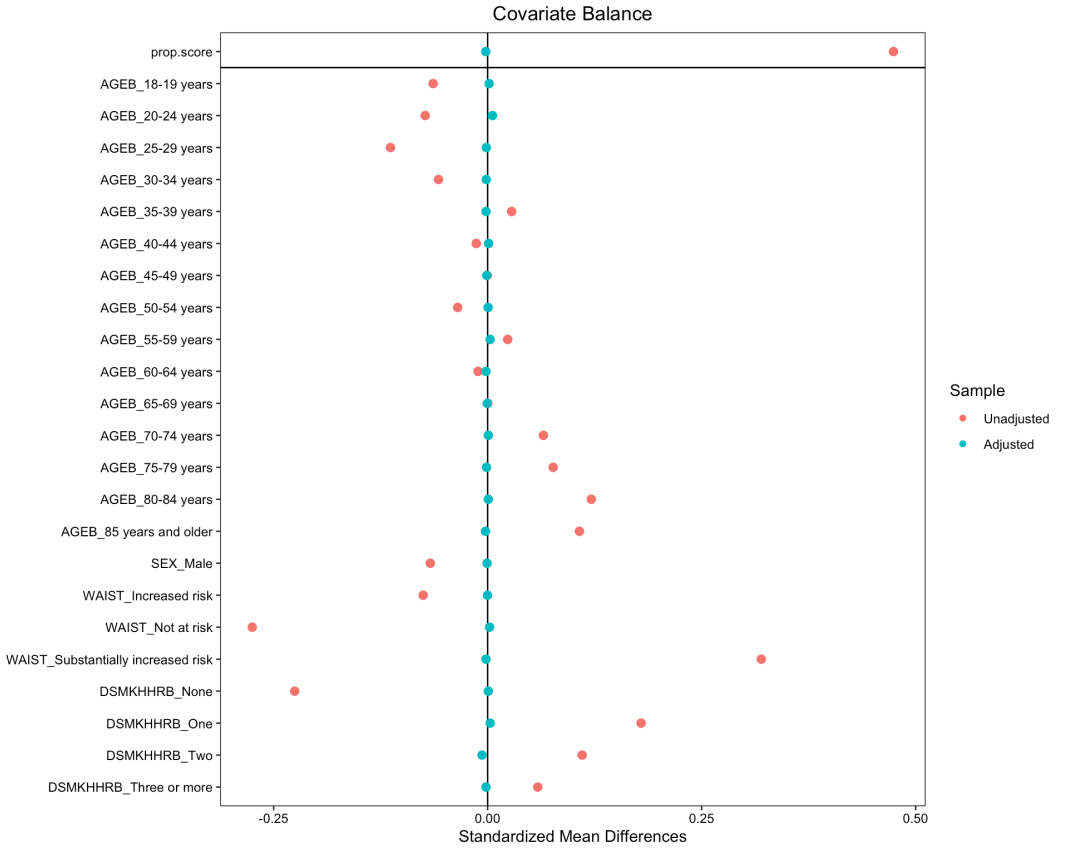
Balance tally for mean differences
                   count
Balanced, <0.1        24
Not Balanced, >0.1     0

Variable with the greatest mean difference
     Variable Diff.Adj    M.Threshold
 DSMKHHRB_Two  -0.0065 Balanced, <0.1

Effective sample sizes
           Met recommended guidelines Did not meet recommended
guidelines
Unadjusted                    2773.
2567.
Adjusted                      2610.42
2416.28
```

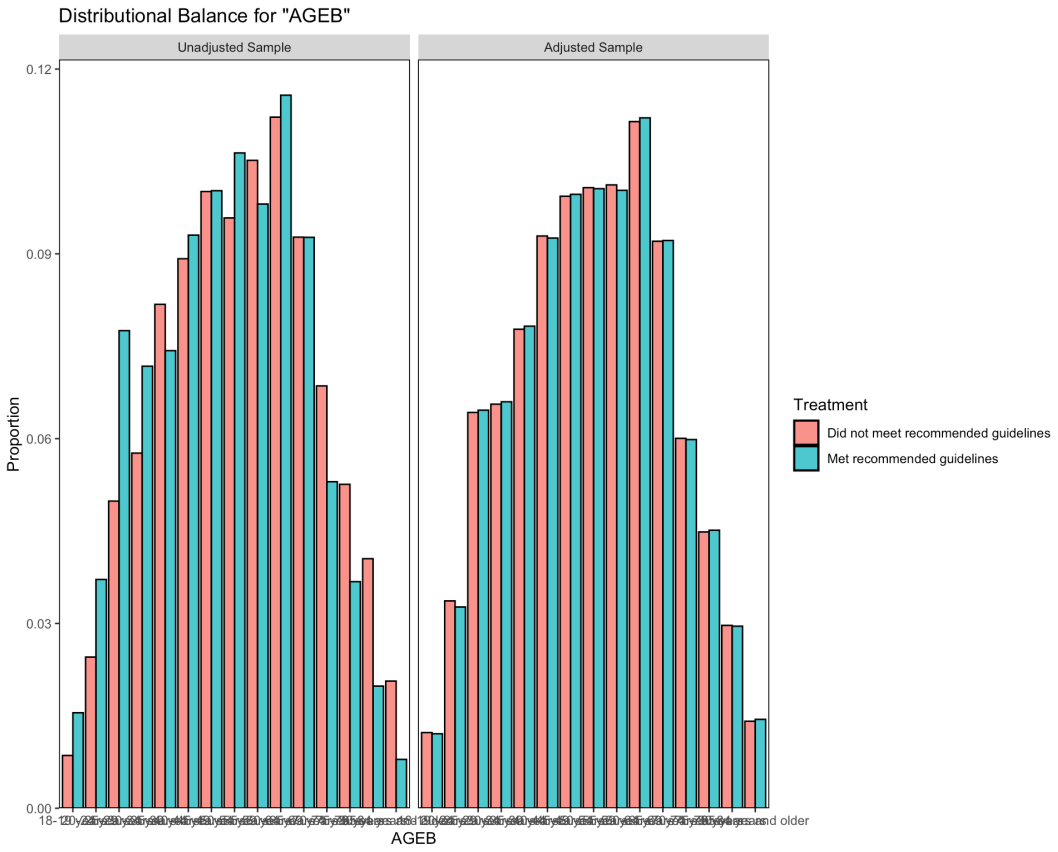
▼ Code

```
love.plot(wmultinom.alt)
```



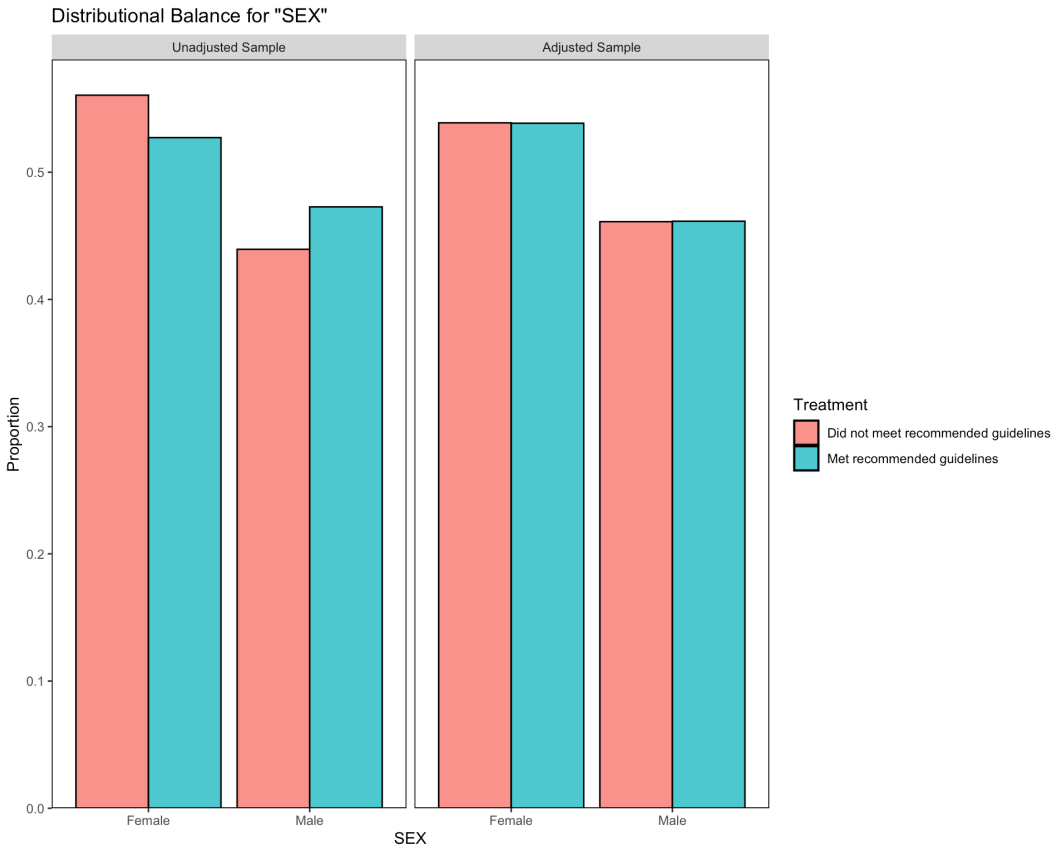
▼ Code

```
# Overlap/distributional balance for each variables
bal.plot(wmultinom.alt, var.name = "AGEB", which = "both", type
```



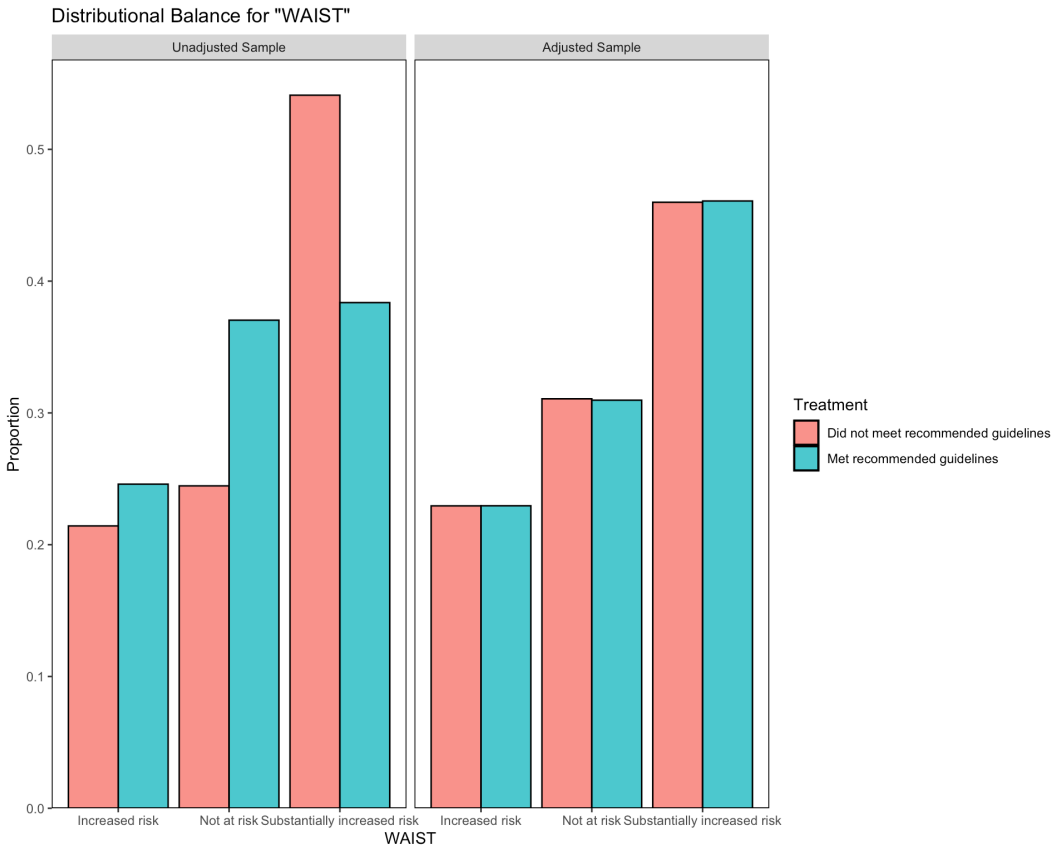
▼ Code

```
bal.plot(wmultinom.alt, var.name = "SEX", which = "both", type =
```



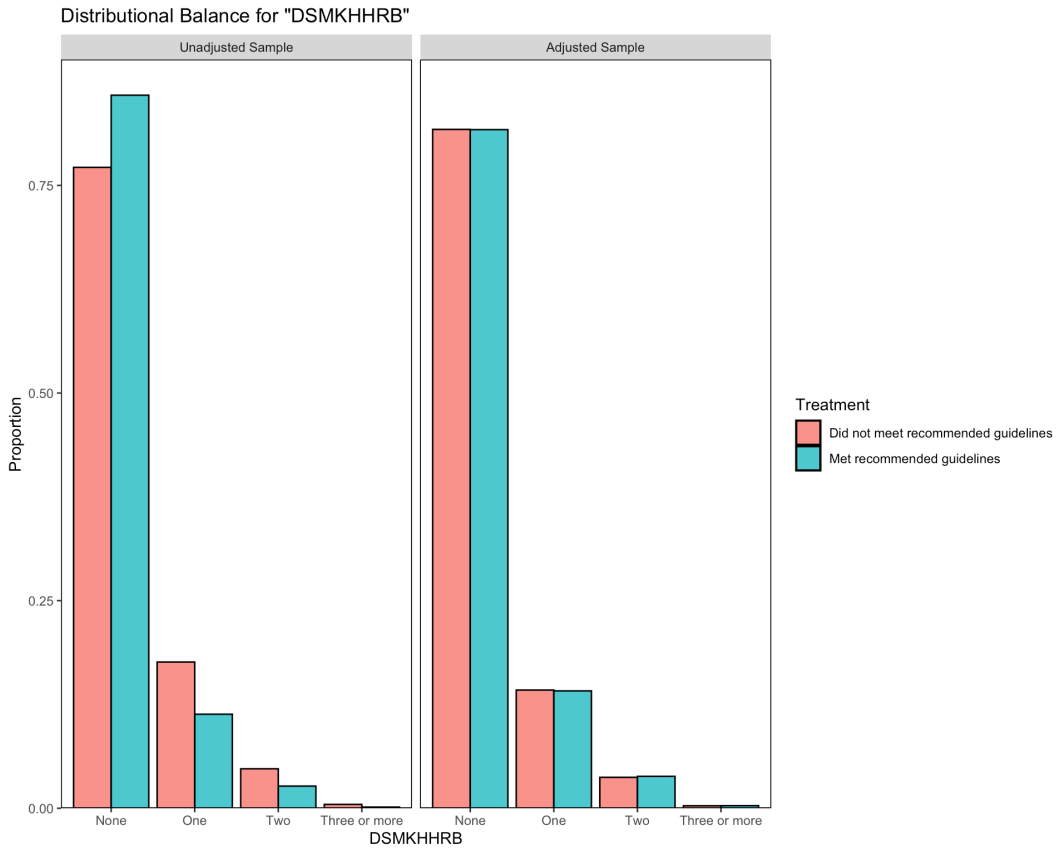
▼ Code

```
bal.plot(wmultinom.alt, var.name = "WAIST", which = "both", type
```



▼ Code

```
bal.plot(wmultinom.alt, var.name = "DSMKHHRB", which = "both", t
```



▼ Code

```
# Extreme weights; high values suggest we should try to trim wei
summary(wmultinom.alt)
```

```
                Summary of weights

- Weight ranges:

          Min                                  Max
treated 1.181 |-------------|               3.78
control 1.36   |------------------------| 6.518

- Units with the 5 most extreme weights by group:

          2574  2206  1742   430   304
 treated  3.78  3.78  3.78  3.78  3.78
           631   959   782  3688  2920
 control 4.196 4.335 5.151 5.425 6.518

- Weight statistics:

        Coef of Var   MAD Entropy # Zeros
treated        0.25 0.198   0.03        0
```

```
control        0.25 0.183   0.028       0

- Effective Sample Sizes:

           Control Treated
Unweighted 2773.   2567.
Weighted   2610.42 2416.28
```

▼ Code

```
trimmed.wmultinom.alt = trim(wmultinom.alt, at = .99)
summary(trimmed.wmultinom.alt)
```

```
                Summary of weights

- Weight ranges:

          Min                                  Max
treated 1.181 |--------------------------| 3.609
control 1.36    |------------------------| 3.609

- Units with the 5 most extreme weights by group:

           509   430   420   304    37
 treated 3.609 3.609 3.609 3.609 3.609
           980   959   782   631   526
 control 3.609 3.609 3.609 3.609 3.609

- Weight statistics:

        Coef of Var   MAD Entropy # Zeros
treated       0.249 0.198   0.03        0
control       0.239 0.181   0.026       0

- Effective Sample Sizes:

           Control Treated
Unweighted 2773.   2567.
Weighted   2623.34 2417.55
```

GGT

▼ Code

```
# Formula for GGT; use full population (and complete cases only)
fm.ggt = EXREGUIC ~ AGEB + SEX + WAIST + DSMKHHRB
```

```
fdat.comp = fdat.full %>% drop_na()

# WeightIt
wmultinom.ggt = weightit(fm.ggt, fdat.comp, method = "glm", est
set.cobalt.options(binary = "std")

# Check SMD; generally < 0.1 is considered well-balanced. Also
bal.tab(wmultinom.ggt, thresholds = c(m = .1))
```

```
Balance Measures
                                     Type Diff.Adj
M.Threshold
prop.score                       Distance  -0.0022 Balanced,
<0.1
AGEB_18-19 years                   Binary   0.0018 Balanced,
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AGEB_40-44 years                   Binary   0.0012 Balanced,
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AGEB_45-49 years                   Binary  -0.0010 Balanced,
<0.1
AGEB_50-54 years                   Binary   0.0006 Balanced,
<0.1
AGEB_55-59 years                   Binary   0.0029 Balanced,
<0.1
AGEB_60-64 years                   Binary  -0.0019 Balanced,
<0.1
AGEB_65-69 years                   Binary  -0.0004 Balanced,
<0.1
AGEB_70-74 years                   Binary   0.0008 Balanced,
<0.1
AGEB_75-79 years                   Binary  -0.0013 Balanced,
<0.1
AGEB_80-84 years                   Binary   0.0007 Balanced,
<0.1
AGEB_85 years and older            Binary  -0.0026 Balanced,
<0.1
SEX_Male                           Binary  -0.0007 Balanced,
<0.1
WAIST_Increased risk               Binary  -0.0001 Balanced,
```

```
<0.1
WAIST_Not at risk                    Binary   0.0023 Balanced,
<0.1
WAIST_Substantially increased risk   Binary  -0.0020 Balanced,
<0.1
DSMKHHRB_None                        Binary   0.0008 Balanced,
<0.1
DSMKHHRB_One                         Binary   0.0030 Balanced,
<0.1
DSMKHHRB_Two                         Binary  -0.0065 Balanced,
<0.1
DSMKHHRB_Three or more               Binary  -0.0020 Balanced,
<0.1

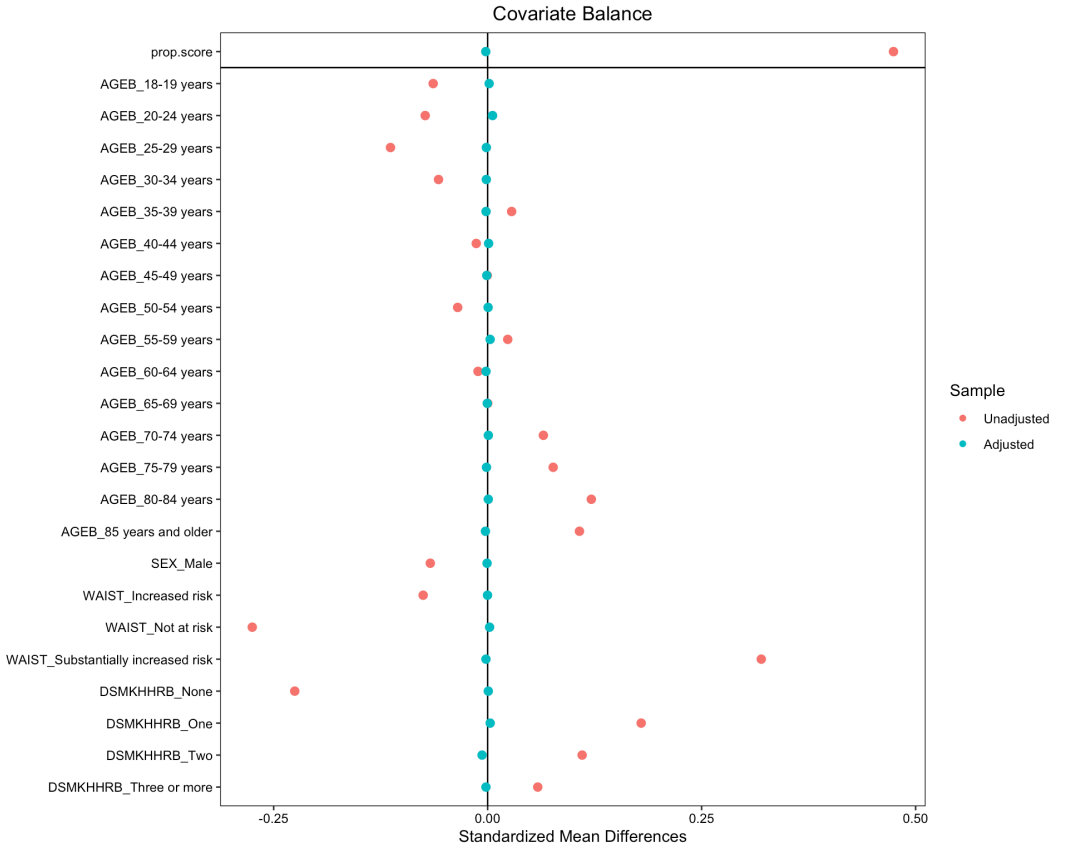
Balance tally for mean differences
                   count
Balanced, <0.1        24
Not Balanced, >0.1     0

Variable with the greatest mean difference
     Variable Diff.Adj    M.Threshold
 DSMKHHRB_Two  -0.0065 Balanced, <0.1

Effective sample sizes
           Met recommended guidelines Did not meet recommended
guidelines
Unadjusted                    2773.
2567.
Adjusted                      2610.42
2416.28
```

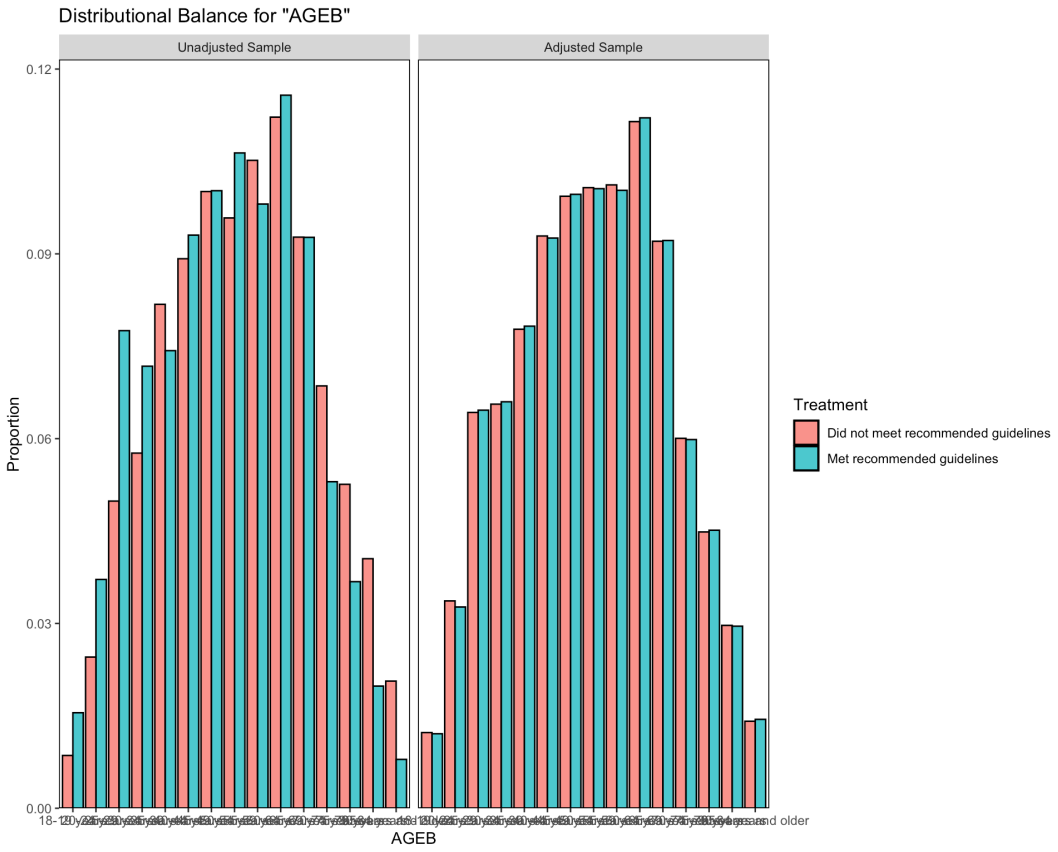
▼ Code

```
love.plot(wmultinom.ggt)
```



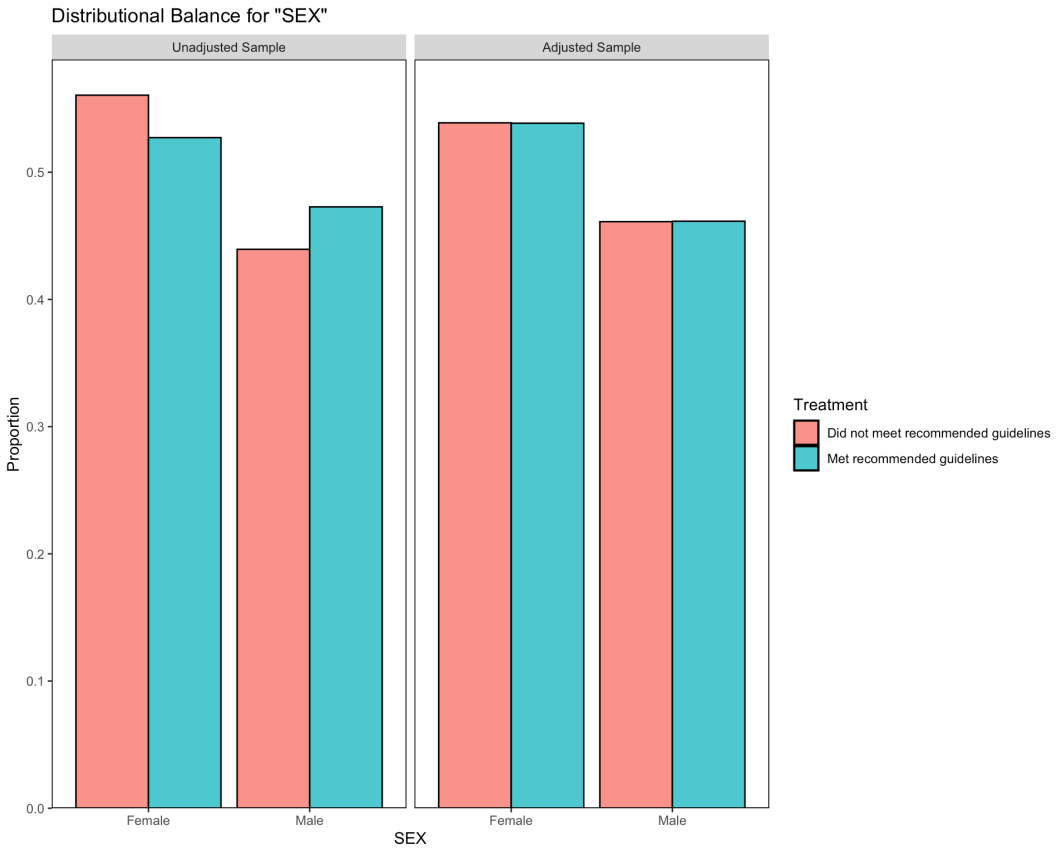
▼ Code

```
# Overlap/distributional balance for each variables
bal.plot(wmultinom.ggt, var.name = "AGEB", which = "both", type
```



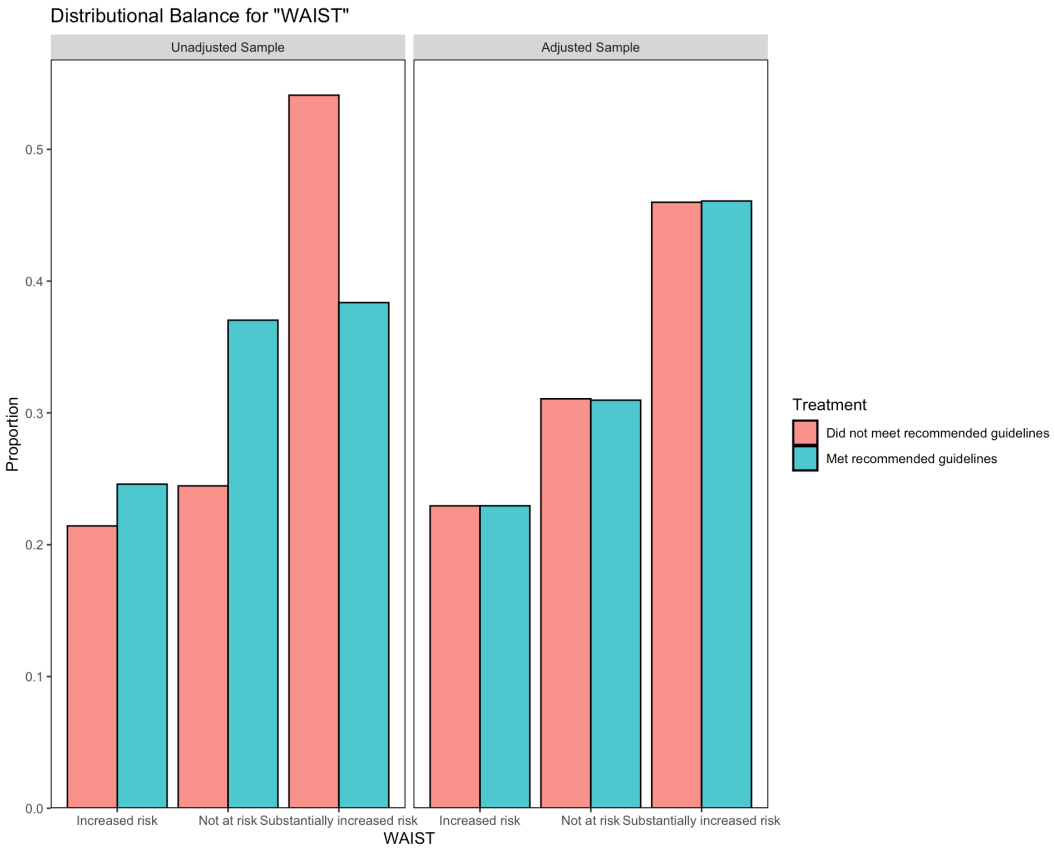
▼ Code

```
bal.plot(wmultinom.ggt, var.name = "SEX", which = "both", type =
```



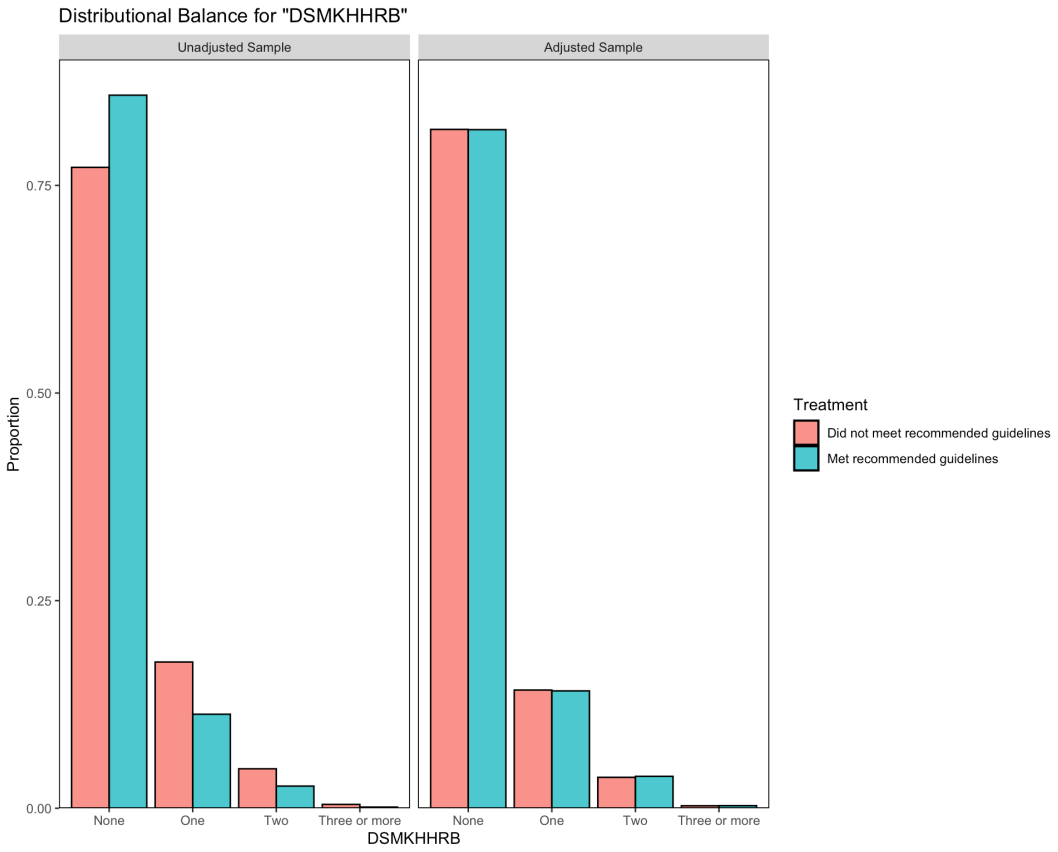
▼ Code

```
bal.plot(wmultinom.ggt, var.name = "WAIST", which = "both", type
```



▼ Code

```
bal.plot(wmultinom.ggt, var.name = "DSMKHHRB", which = "both", t
```



▼ Code

```
# Extreme weights
summary(wmultinom.ggt)
```

```
                Summary of weights

- Weight ranges:

          Min                                  Max
treated 1.181 |-------------|              3.78
control 1.36   |-------------------------| 6.518

- Units with the 5 most extreme weights by group:

          2574  2206  1742   430   304
 treated  3.78  3.78  3.78  3.78  3.78
           631   959   782  3688  2920
 control 4.196 4.335 5.151 5.425 6.518

- Weight statistics:

        Coef of Var   MAD Entropy # Zeros
treated        0.25 0.198   0.03        0
control        0.25 0.183   0.028       0
```

```
- Effective Sample Sizes:

           Control Treated
Unweighted 2773.   2567.
Weighted   2610.42 2416.28
```

▼ Code

```
trimmed.wmultinom.ggt = trim(wmultinom.ggt, at = 0.99)
summary(trimmed.wmultinom.ggt)
```

```
                Summary of weights

- Weight ranges:

          Min                                  Max
treated 1.181 |--------------------------| 3.609
control 1.36    |----------------------| 3.609

- Units with the 5 most extreme weights by group:

           509   430   420   304    37
 treated 3.609 3.609 3.609 3.609 3.609
           980   959   782   631   526
 control 3.609 3.609 3.609 3.609 3.609

- Weight statistics:

        Coef of Var   MAD Entropy # Zeros
treated       0.249 0.198   0.03        0
control       0.239 0.181   0.026       0

- Effective Sample Sizes:

           Control Treated
Unweighted 2773.   2567.
Weighted   2623.34 2417.55
```

The Love plot compares covariate imbalance before and after propensity-score weighting. Successful weighting should move the adjusted standardized mean differences towards zero, with values below 0.1 generally indicating acceptable balance. Improved balance suggests that participants who met and did not meet physical-activity guidelines are more comparable with respect to the measured confounders, reducing their potential to explain the observed exercise–ALT/GGT association.

Extreme inverse-probability weights indicate participants whose observed physical-activity status was very unlikely given their measured covariates. Such weights may reflect limited overlap between exercise groups and can substantially increase variance by allowing a small number of observations to dominate the analysis. We therefore inspect the weight distribution and effective sample size before estimating the weighted effect.

Also, since our trimming changed the summary substantially, this may be an indication of poor overlap.

Estimated Outcome Effects

We attach the weights we found earlier to the model, and estimate the effects.

▼ Code

```
# ALT
alt.fmlr.weighted = multinom_weightit(
  ALT3 ~ EXREGUIC,
  data = fdat.comp,
  weightit = wmultinom.alt
)
avg_comparisons(alt.fmlr.weighted,
                variables = "EXREGUIC",
                type = "probs"
) %>% tidy() %>% select(-c(contrast)) %>%
  kable(caption = "Average Comparisons of ALT Using PSW (Contras
        digits = 3, col.names = c("Term", "Group", "Estimate", "
                       "p-value", "s-value", "CI LB", "CI UB"))
```

Average Comparisons of ALT Using PSW (Contrast = Did Not Meet ~ Met)

Term	Group	Estimate	SE	Statistic	p-value	s-value	CI LB	CI UB
EXREGUIC	Low	-0.014	0.011	-1.237	0.216	2.210	-0.036	0.008
EXREGUIC	Borderline	-0.008	0.009	-0.976	0.329	1.603	-0.025	0.008
EXREGUIC	High	0.022	0.009	2.499	0.012	6.326	0.005	0.040

▼ Code

```
# GGT
ggt.fmlr.weighted = multinom_weightit(
  GGT3 ~ EXREGUIC,
  data = fdat.comp,
```

```
  weightit = wmultinom.ggt
)
avg_comparisons(ggt.fmlr.weighted,
                variables = "EXREGUIC",
                type = "probs") %>% tidy() %>% select(-c(contras
  kable(caption = "Average Comparisons of GGT Using PSW (Contras
        digits = 3, col.names = c("Term", "Group", "Estimate", "
                      "p-value", "s-value", "CI LB", "CI UB"))
```

Average Comparisons of GGT Using PSW (Contrast = Did Not Meet ~ Met)

Term	Group	Estimate	SE	Statistic	p-value	s-value	CI LB	CI UB
EXREGUIC	Low	-0.044	0.011	-4.051	0.000	14.256	-0.065	-0.023
EXREGUIC	Borderline	0.004	0.006	0.557	0.578	0.792	-0.009	0.016
EXREGUIC	High	0.040	0.009	4.281	0.000	15.717	0.022	0.058

We use the same interpretation as before.